Modified Scope

Due to the shortage of available time, we have shifted into making 2 separate circuits and assign it to 2 different teams as follows:

1- **Circuit 1** will be responsible for GPS, Tracking Geofencing, and GSM to communicate with the server.
2- **Circuit 2** will be responsible for controlling the water supply.

Circuit 2 will contain:

- MCU
- Flowmeter
- Electronic Valve
- RFID Reader
- Keypad
- Thermal Printer
- Pushbutton for manual on/off Valve operation
- Finish Pushbutton to end the filling for the client.
- Pushbutton to fetch daily route schedule from the server.

- The data will be transferred to the server upon the from the 2nd circuit through GSM Module on the 1st circuit.
- Circuit 1 will "Activate" circuit 2 upon Geofence Verification.
- Circuit 2 will only be activated within approved Geofence (received from the server).
- Client will verify identity through RFID card or OTP (Both must be included in the 2nd Circuit).
- Daily route schedule is updated in the server and must be manually imported from the server onto the device on board (using the manual pushbutton) with the following data:

Client Name	Client ID	RFID Card Number	Allocated Water	OTP

- Manual On/Off button to bypass automatic operation and for emergency situation.
- Data will be displayed on the vehicle including:
    - Welcome Message upon Client verification.
    - Allocated amount of water
    - Live display of flowrate
    - Overall water amount filled.
    - Goodbye message.
- Finish button will send the data to the server.
- Thermal will print receipt of the operation upon completion of water filling process.